

HSC Practice Questions

Simultaneous Equations

- 1 In a park the only animals are goannas and emus. Let x be the number of goannas and let y be the number of emus.

The number of goannas plus the number of emus in the park is 31. Hence $x + y = 31$.

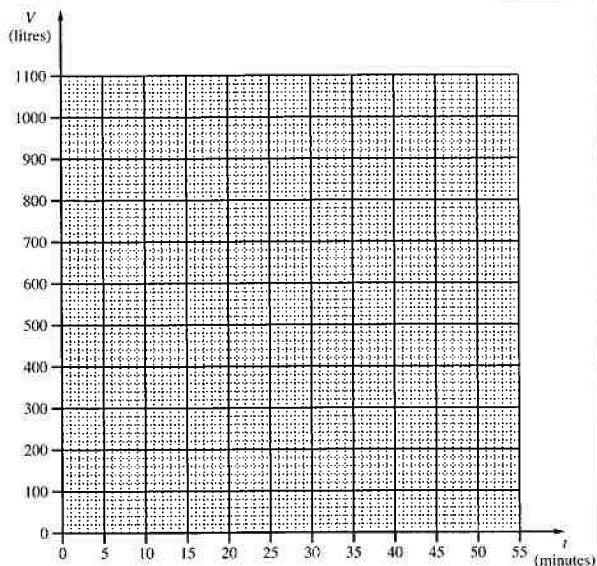
Each goanna has four legs and each emu has two legs. In total the emus and goannas have 76 legs.

By writing another relevant equation and graphing both equations on the grid on the following page, find the number of goannas and the number of emus in the park. (4 marks)

(Q34, 2021 HSC) **Hard**

- 2 There are two tanks on a property, Tank A and Tank B. Initially, Tank A holds 1000 litres of water and Tank B is empty.

- a Tank A begins to lose water at a constant rate of 20 litres per minute. The volume of water in Tank A is modelled by $V = 1000 - 20t$ where V is the volume in litres and t is the time in minutes from when the tank begins to lose water. On the grid below, draw the graph of this model and label it as Tank A. (1 mark) **Easy**



- b Tank B remains empty until $t = 15$ when water is added to it at a constant rate of 30 litres per minute.

By drawing a line on the grid on the previous page, or otherwise, find the value of t when the two tanks contain the same volume of water. (2 marks) **Medium**

- c Using the graphs drawn, or otherwise, find the value of t (where $t > 0$) when the total volume of water in the two tanks is 1000 litres. (1 mark) **Hard**

CO (Q24, 2020 HSC)

- 3 Last Saturday, Luke had 165 followers on social media. Rhys had 537 followers. On average, Luke gains another 3 followers per day and Rhys loses 2 followers per day. If x represents the number of days since last Saturday and y represents the number of followers, which pair of equations model this situation?

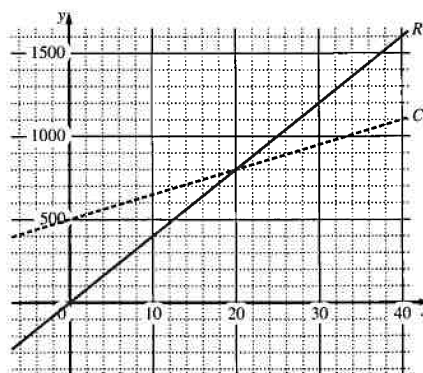
- A Luke: $y = 165x + 3$ B Luke: $y = 165 + 3x$
Rhys: $y = 537x - 2$ Rhys: $y = 537 - 2x$
C Luke: $y = 3x + 165$ D Luke: $y = 3 + 165x$
Rhys: $y = 2x - 537$ Rhys: $y = 2 - 537x$

(1 mark)

(Q14, 2019 HSC) **Easy**

- 4 A small business makes and sells bird houses.

Technology was used to draw straight-line graphs to represent the cost of making bird houses (C) and the revenue from selling bird houses (R). The x -axis displays the number of bird houses and the y -axis displays the cost/ revenue in dollars.



- a How many bird houses need to be sold to break even? (1 mark) **Easy**
b By first forming equations for cost (C) and revenue (R), determine how many bird houses need to be sold to earn a profit of \$1900. (3 marks) **Hard**

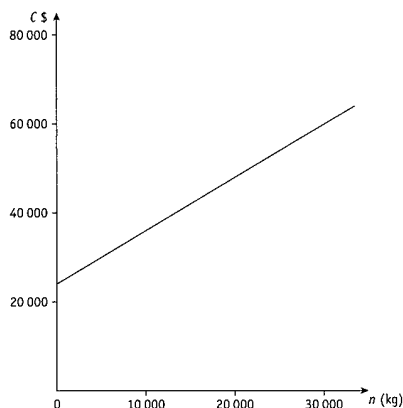
(Q36, 2019 HSC)

- 5 The cost, C , in dollars, of producing n kilograms of potatoes is given by $C = 1.2n + 24000$. The revenue from selling n kilograms of potatoes is given by $R = 2.4n$.

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The cost, C , for the production of n kilograms of potatoes is graphed below.



- On the graph above draw the graph for the equation $R = 2.4n$ to identify the break-even point. (1 mark) **Medium**
- What profit will be made if 30 000 kg of potatoes are grown and sold? (2 marks) **Medium**
- If a loss of \$ t occurs, write an expression to represent the mass of potatoes grown and sold. (1 mark) **Hard**

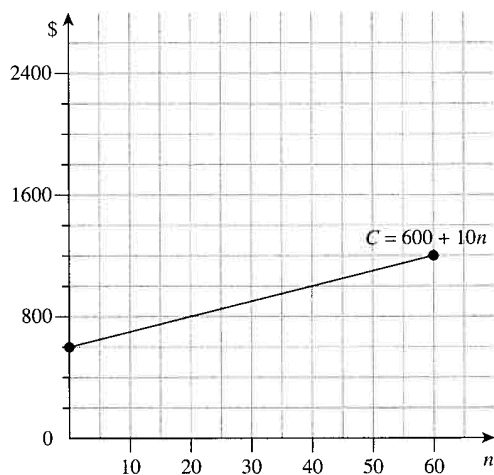
Bonus question (see page iv)

- A meal delivery business produces and sells dinner packages.

The cost \$ C to make and deliver n dinner packages is modelled by the equation $C = 600 + 10n$.

The revenue \$ R is the amount of money received when selling n dinner packages.

The diagram below shows the graph of $C = 600 + 10n$, where the horizontal axis represents the number of packages and the vertical axis displays the cost, or revenue, in dollars.

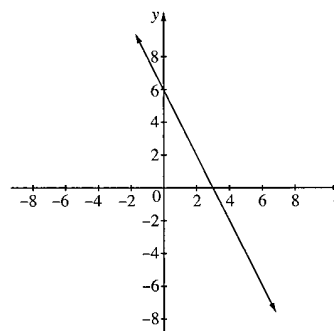


It is known that the company breaks even at 20 dinner packages.

- On the above diagram draw a straight line representing the graph of R . (1 mark) **Medium**
- If the equation for R is written in the form $R = kn$, what is the value of k ? (1 mark) **Medium**
- How many dinner packages need to be sold for the company to make a profit of \$900? (2 marks) **Medium**

Bonus question

- The graph of the line with equation $y = 6 - 2x$ is shown.



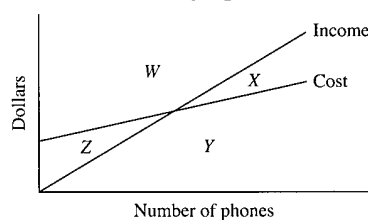
When the graph of the line with equation $y = x + 3$ is also drawn on this number plane, what will be the point of intersection of the two lines?

- | | |
|-----------------|-----------------|
| A (0, 6) | B (1, 4) |
| C (2, 2) | D (3, 0) |

(1 mark)

(Q17, 2017 HSC) **Medium**

- A company manufactures phones. The company's income equation and cost equation are drawn on the same graph.



Which region of the graph is the profit zone?

- | | |
|------------|------------|
| A W | B X |
| C Y | D Z |

(1 mark)

(Q4, 2016 HSC) **Easy**

- A function centre hosts events for up to 500 people. The cost C , in dollars, for the centre to host an event, where x people attend, is given by:

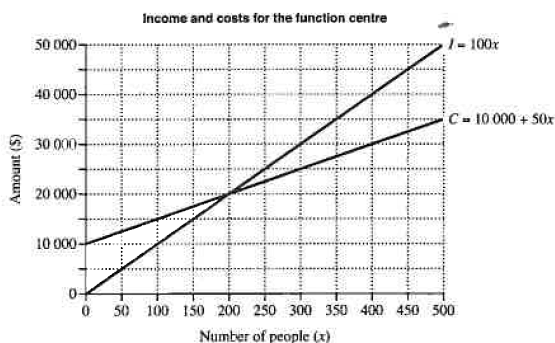
$$C = 10000 + 50x$$

The centre charges \$100 per person. Its income I , in dollars, is given by:

$$I = 100x$$

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How much greater is the income of the function centre when 500 people attend an event, than its income at the breakeven point?

- A \$15 000 B \$20 000
C \$30 000 D \$40 000 (1 mark)

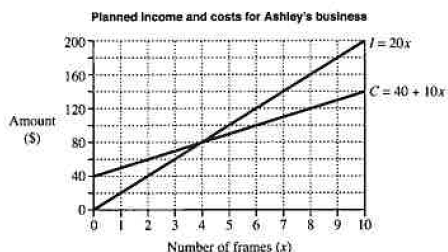
(Q20, 2011 HSC) Medium

- 10 Ashley makes picture frames as part of her business. To calculate the cost, C , in dollars, of making x frames, she uses the equation

$$C = 40 + 10x.$$

She sells the frames for \$20 each and determines her income, I , in dollars, using the equation

$$I = 20x.$$



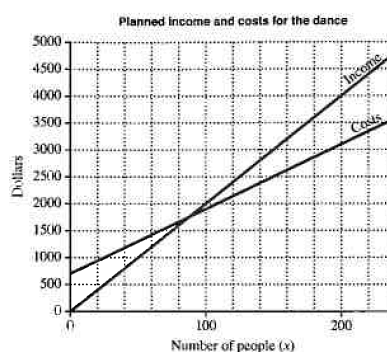
Use the graph to solve the two equations simultaneously for x and explain the significance of this solution for Ashley's business. (2 marks)

(Q24b, 2010 HSC) Medium

- 11 Sue and Mikey are planning a fund-raising dance. They can hire a hall for \$400 and a band for \$300. Refreshments will cost them \$12 per person.

- i Write a formula for the cost (\$ C) of running the dance for x people. (1 mark) Easy

The graph shows planned income and costs when the ticket price is \$20.

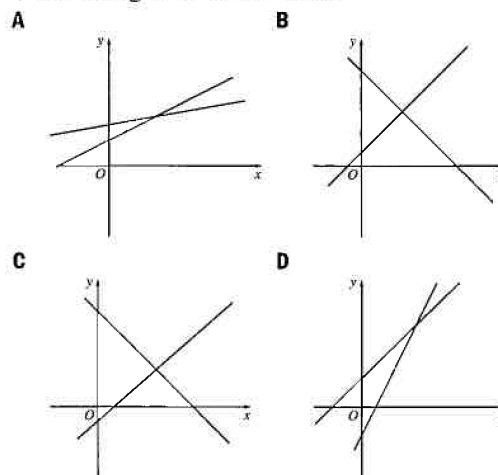


- ii Estimate the minimum number of people needed at the dance to cover the costs. (1 mark) Easy
iii How much profit will be made if 150 people attend the dance? (1 mark) Medium
iv Sue and Mikey plan to sell 200 tickets. They want to make a profit of \$1500. What should be the price of a ticket, assuming all 200 tickets will be sold? (3 marks) Hard

(Q28b, 2005 HSC)

- 12 George drew a correct diagram that gave the solution to the simultaneous equations $y = 2x - 5$ and $y = x + 6$.

Which diagram did he draw?



(1 mark)

(Q16, 2004 HSC) Medium

- 13 In 2001, when Toby was in Year 9, he started earning money by juggling at children's parties. He charged \$50 per party.

- i Write a formula for the amount, Q (in dollars), that Toby had earned in 2001 after he had juggled at n parties. (1 mark) Medium
ii By the end of 2001 Toby had juggled at 30 parties.

Draw the straight line graph of Q against n , with n on the horizontal axis and Q on the vertical axis. Use your ruler to draw the axes, and mark a scale on each axis. (3 marks) Hard

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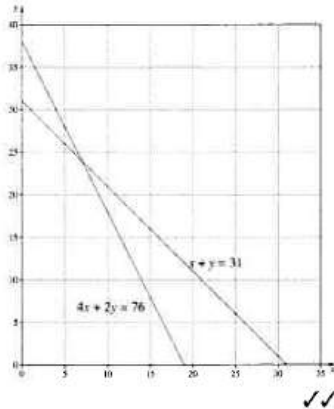
- iii Before Toby started juggling at parties he spent \$300 on juggling equipment. On your graph in part ii draw a horizontal line through the point on the vertical axis where $Q = 300$. Give an interpretation of the point at which this horizontal line crosses the straight line graph of Q against n .

(1 mark)

Hard

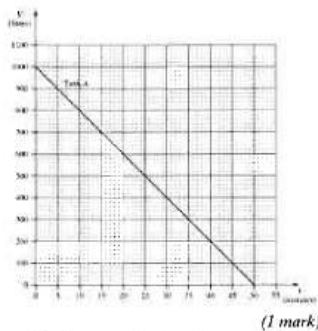
(Q28b, 2002 HSC)

- 1 Considering $x + y = 31$:
When $x = 0$, $y = 31$.
When $y = 0$, $x = 31$.
The total number of legs is 4 for each goanna plus 2 for each emu.
So $4x + 2y = 76$.
When $x = 0$, $2y = 76$
 $y = 38$
When $y = 0$, $4x = 76$
 $x = 19$

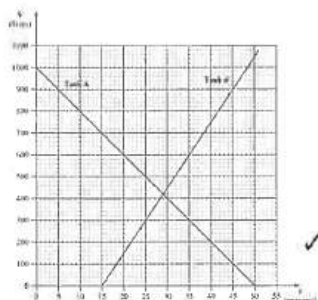


The point of intersection is when $x = 7$ and $y = 24$.
So, from the graph, there are 7 goannas and 24 emus. (4 marks)

- 2 a $V = 1000 - 20t$
Vertical intercept = 1000
Gradient = -20



- b Tank B: $V = 0$ when $t = 15$.
Gradient = 30



The two tanks have the same volume when $t = 29$. (2 marks)

- c From the graph the total volume of the two tanks will be 1000 L when $t = 45$.
(100 L in tank A and 900 L in tank B.) (1 mark)

- 3 Luke has 165 followers plus 3 per day.
So for Luke $y = 165 + 3x$
Rhys has 537 followers less 2 per day.
So for Rhys $y = 537 - 2x$
Answer B

- 4 a 20 bird houses need to be sold to break even. (1 mark)

b Revenue:
y-intercept = 0
Gradient = $\frac{800}{20}$
 $= 40$

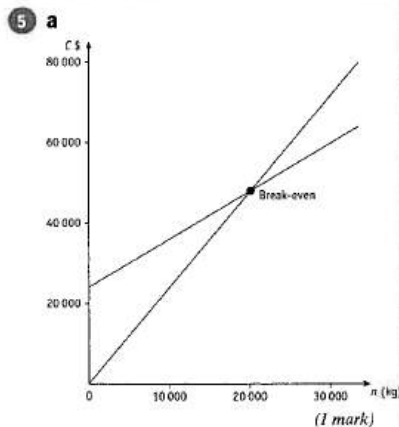
$R: y = 40x$
Cost: y-intercept = 500

Gradient = $\frac{300}{20}$
 $= 15$

$C: y = 15x + 500$
Now profit = revenue - cost
 $= 40x - (15x + 500)$
 $= 40x - 15x - 500$
 $= 25x - 500$

When the profit is \$1900,
 $25x - 500 = 1900$
 $25x = 2400$
 $x = 96$

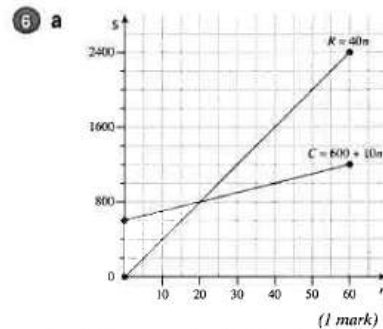
So 96 bird houses need to be sold to earn a profit of \$1900. (3 marks)



- b Substitute 30000:
 $R = 2.4(30000)$
 $= 72000$
 $C = 1.2(30000) + 24000$
 $= 60000$

Profit = $72000 - 60000$
 $= 12000$
 $\therefore$ a profit of \$12 000 (2 marks)

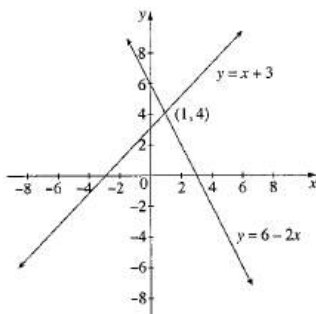
- c Loss = $C - R$
 $t = 1.2n + 24000 - 2.4n$
 $= 24000 - 1.2n$
 $1.2n = 24000 - t$
 $n = \frac{24000 - t}{1.2}$ (1 mark)



- b The equation is $R = kn$.
As the straight line passes through (20, 800), substitute $n = 20$ and $R = 800$ into equation:
 $800 = k(20)$
 $20k = 800$
 $k = 40$ (1 mark)

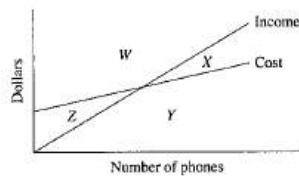
- c The cost is $C = 600 + 10n$ and the revenue is $R = 40n$.
As Profit = Revenue - Cost,
 $= 40n - (600 + 10n)$
 $= 40n - 600 - 10n$
 $= 30n - 600$
Substituting Profit = 900:
 $900 = 30n - 600$
 $30n = 900 + 600$
 $30n = 1500$
 $n = 50$
The company needs to sell 50 dinner packages. (2 marks)

- 7 At the point of intersection:
 $x + 3 = 6 - 2x$
 $3x + 3 = 6$
 $3x = 3$
 $x = 1$
When $x = 1$, $y = 1 + 3 = 4$.
The point of intersection is (1, 4).

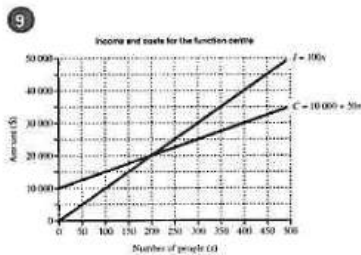


Answer B

- 8 The profit zone is X. [Where the income is greater than the cost]



Answer B



When 500 people attend, the income is \$50,000.

At the breakeven point, the income is \$20,000.

$$\text{Difference} = \$50\,000 - \$20\,000 = \$30\,000$$

Answer C

- 10 The lines intersect when $x = 4$.
So the simultaneous solution of $I = 20x$ and $C = 40 + 10x$ is when $x = 4$.
This is the break-even point for Ashley's business. If she makes 4 frames her costs will be \$80 and if she sells 4 frames her income will also be \$80. If she makes and sells more than 4 frames she will make a profit, but if she sells fewer than 4 frames she will make a loss.

(2 marks)

- 11 i $C = \$400 + \$300 + \$12 \times x$
 $= \$700 + \$12x$

(1 mark)

- ii The two lines drawn, meet at approximately $x = 90$.
 $\therefore$ About 90 people will need to attend.

(1 mark)

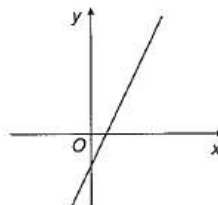
- iii If $x = 150$,
 $C = \$700 + \12×150
 $= \$2500$
Income = \$3000 (from graph)
 $\therefore$ Profit = \$3000 - \$2500
 $= \$500$

(1 mark)

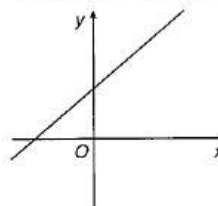
- iv If $x = 200$,
 $C = \$700 + \12×200
 $= \$3100$
Profit = \$1500
Now Income = Cost + Profit
 $= \$3100 + \1500
 $= \$4600$
 $\therefore$ Price of ticket
 $= \$4600 \div 200$
 $= \$23$

(3 marks)

- 12 $y = 2x - 5$ has a positive gradient of 2 and a y-intercept of -5:



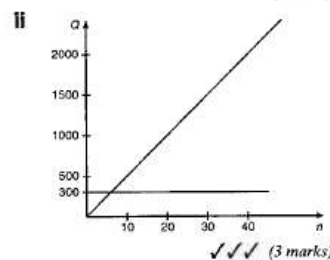
$y = x + 6$ has a positive gradient of 1 and a y-intercept of 6:



Answer D

- 13 i $Q = 50n$

(1 mark)



✓✓✓ (3 marks)

- iii Where the two lines intersect is the breakeven point. Until he has done 6 parties, Toby has not made a profit.

(1 mark)